

THE MEDIUM ACCESS CONTROL SUBLAYER

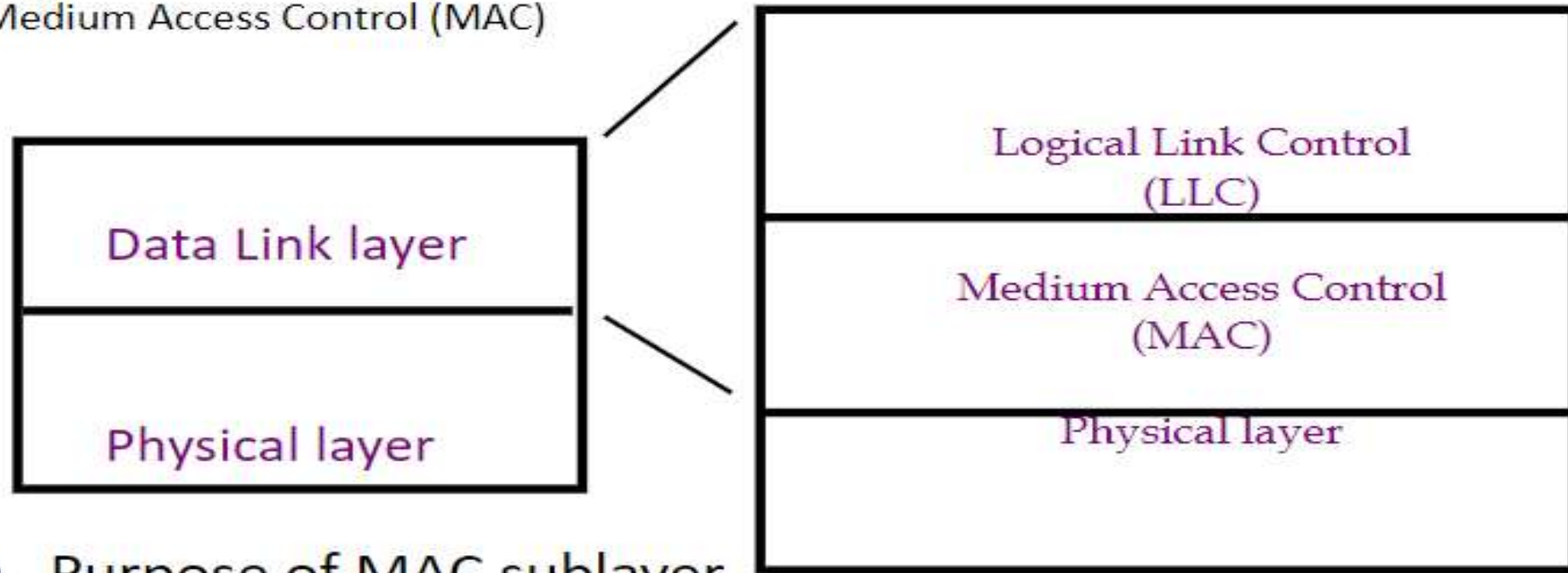
4.1 THE CHANNEL ALLOCATION PROBLEM

4.2 MULTIPLE ACCESS PROTOCOLS

Data Link Layer - Medium Access Control

Data Link Layer consists of two sublayers:

1. Logical Link Control (LLC)
2. Medium Access Control (MAC)



- Purpose of MAC sublayer
 - control access to transmission medium turning shared channel into intermittent point-to-point bit pipe (channel allocation)

4.1 THE CHANNEL ALLOCATION PROBLEM

- **THE CHANNEL ALLOCATION :** how to allocate a single broadcast channel among competing users.
- The channel might be :
 - a portion of the wireless spectrum in a geographic region,
 - or a single wire or optical fiber to which multiple nodes are connected.
- **Types of The channel allocation:**
 - 1.Static Channel Allocation,
 - 2.Multible Access protocols

4.1.1 Static Channel Allocation

- The traditional way of channel allocation is FDM scheme (Frequency Division Multiplexing).
- FDM: If there are N users, the bandwidth is divided into N equal-sized portions (frequency band), with each user being assigned one portion.

4.1.1 Static Channel Allocation(3)

when the number of senders is large and varying or the traffic is bursty, FDM presents some problems:

- If (number of spectrum portions < number of stations)
 - » some of them will be denied permission for lack of bandwidth.
- If (number of spectrum portions > number of stations)
 - » a large piece of valuable spectrum will be wasted.
- when some stations are quiescent,
 - » their bandwidth is simply_wasted_. They are not using it, and no one else is allowed to use it either.

4.1.1 Static Channel Allocation(4)

- data traffic is extremely bursty, often with peak traffic to mean traffic ratios of 1000:1. Consequently, most of the channels will be idle most of the time.

4.2 MULTIPLE ACCESS PROTOCOLS

- **MULTIPLE ACCESS** : enabling many devices to gain access to the channel at the same time.
- Multiple Access Protocols:
 - Contention protocols
 - ALOHA protocols
 - Carrier Sense Multiple Access (CSMA) protocols
 - Collision-Free Protocols
 - Limited-Contention Protocols

4.2.1 ALOHA Protocols

- **Pure ALOHA**: let stations transmit whenever they have a frame to be sent.
- **Slotted ALOHA**:
 - Time is divided into discrete slots (i.e. slot is interval corresponding to one frame).
 - And a station that has a frame to be sent is required to wait for the beginning of the next slot.
 - this algorithm leads to better channel utilization but longer delays than pure ALOHA

4.2.1 ALOHA Protocols (1)

- in ALOHA, If a frame was collided (collision happened), the sender just waits a random amount of time and sends it again. The waiting time must be random or the same frames will collide again.

Pure ALOHA

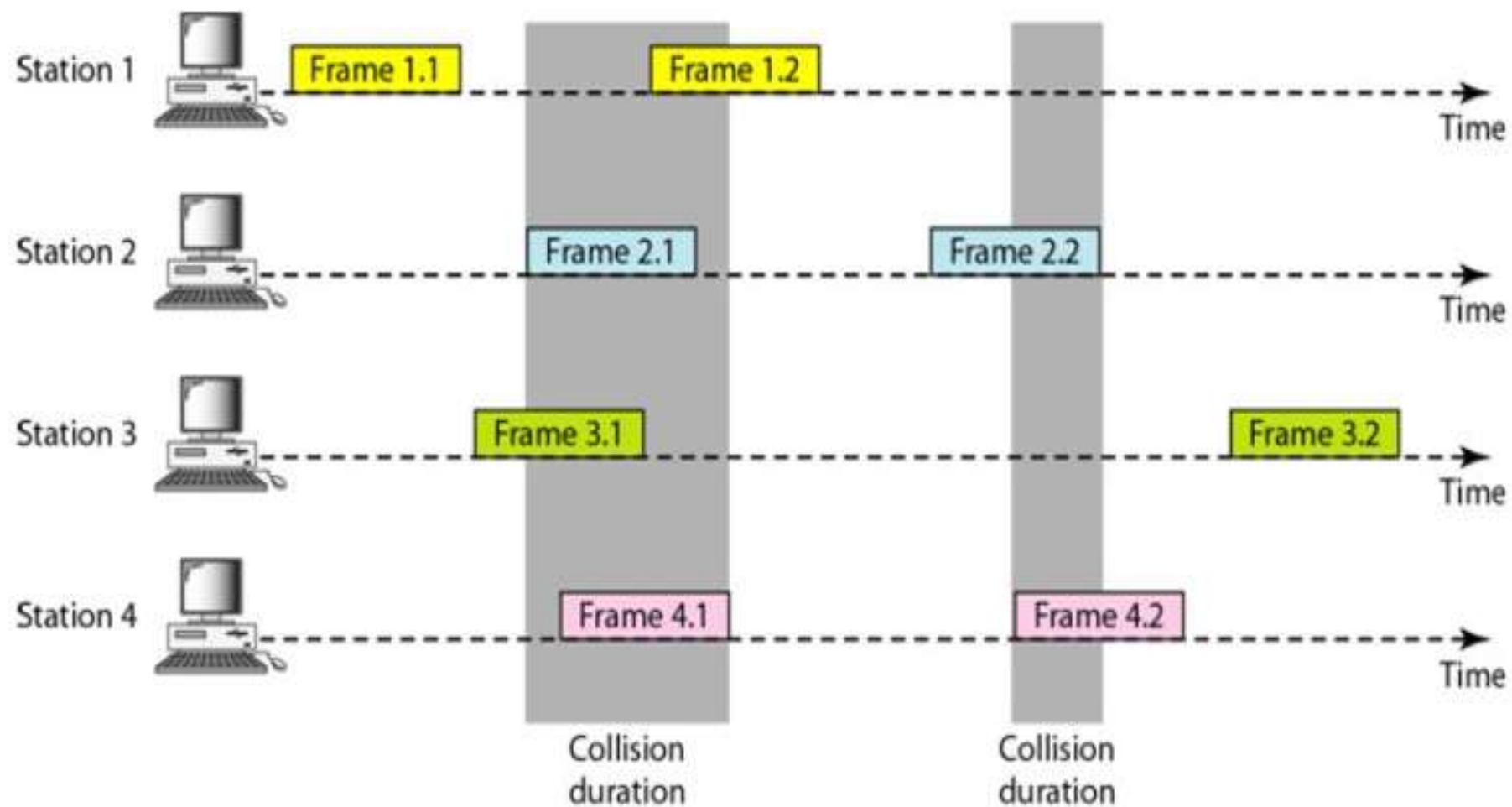
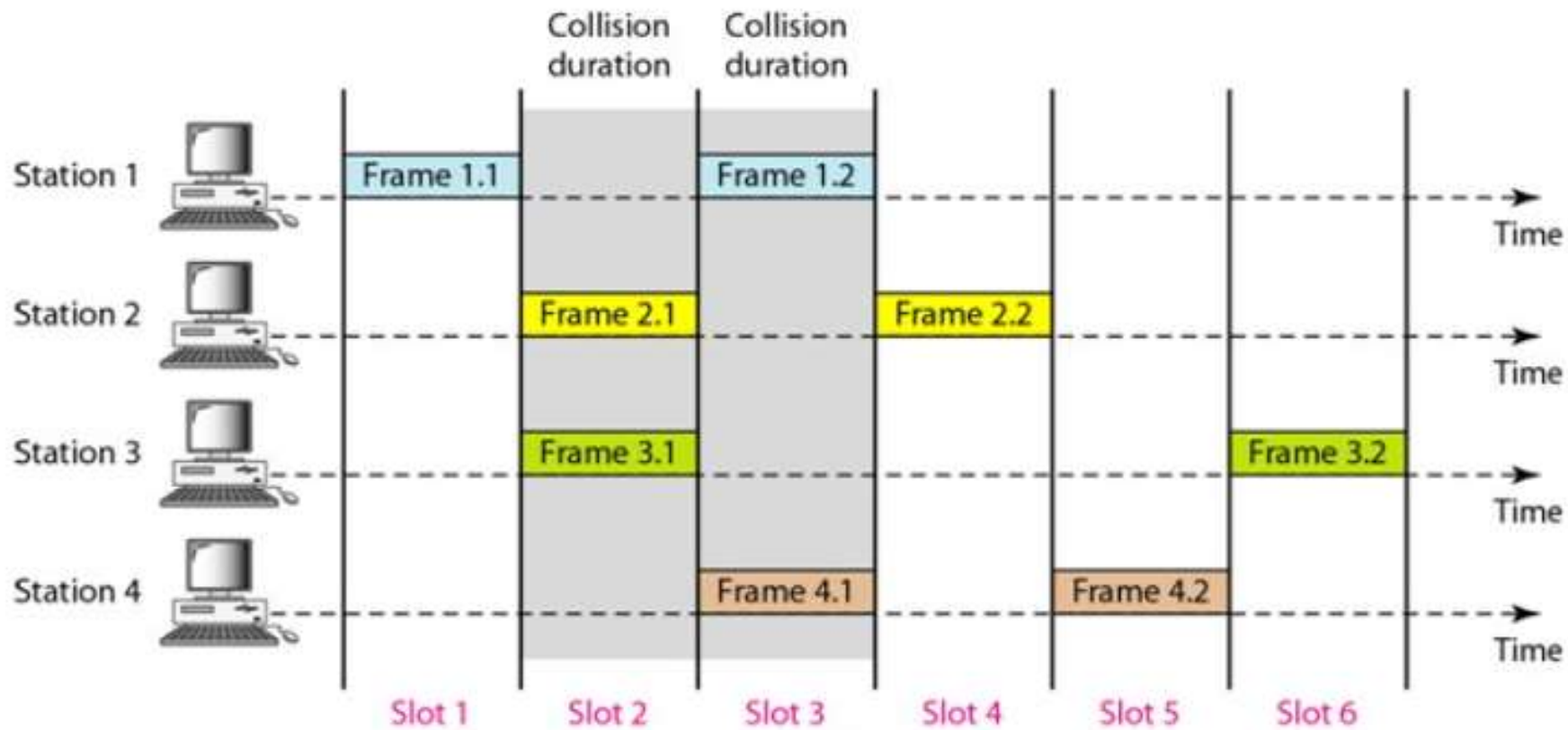


Figure 4-1. In pure ALOHA, frames are transmitted at completely arbitrary times.

Slotted ALOHA



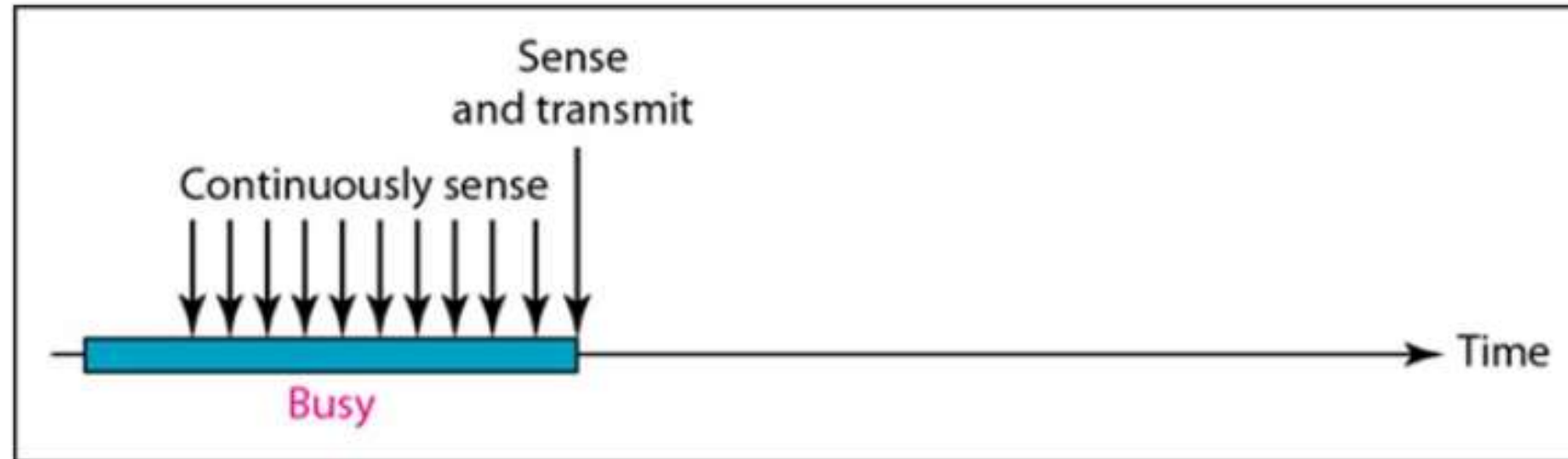
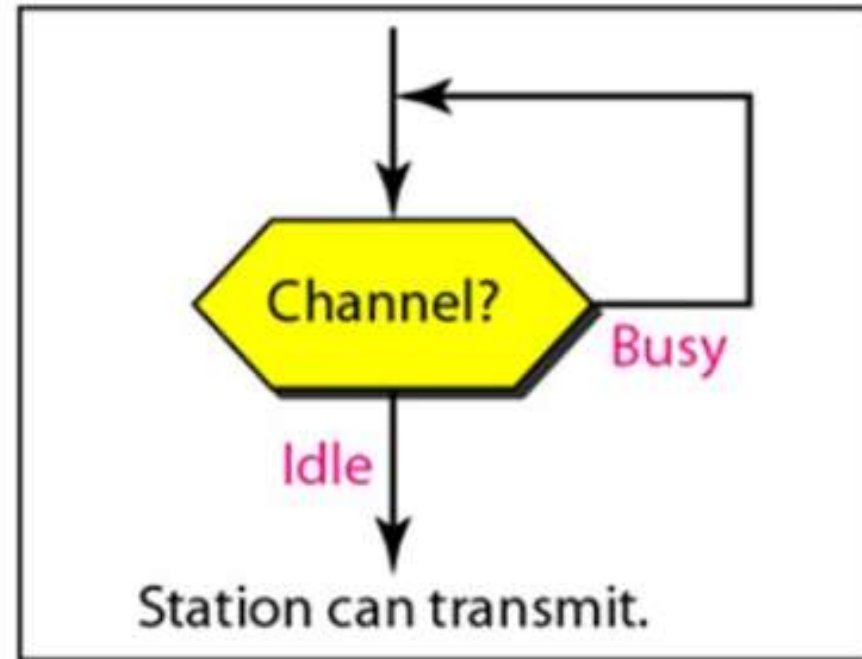
4.2.2 Carrier Sense Multiple Access Protocols (CSMA)

- **CSMA:** Protocols in which stations listen for a carrier and act accordingly
- CSMA protocols can achieve a much better utilization than ALLOHA protocols.
- Versions of CSMA:
 - 1-persistent CSMA
 - nonpersistent CSMA
 - CSMA with Collision Detection

1-persistent CSMA

- When a station has a frame to send, it first listens to the channel (idle or busy)
 - If the channel is idle, the station sends its frame.
 - if the channel is busy, the station continuously listen to to the channel until it becomes idle. Then the station transmits a frame.
 - If a collision occurs, the station waits a random amount of time and starts all over again.

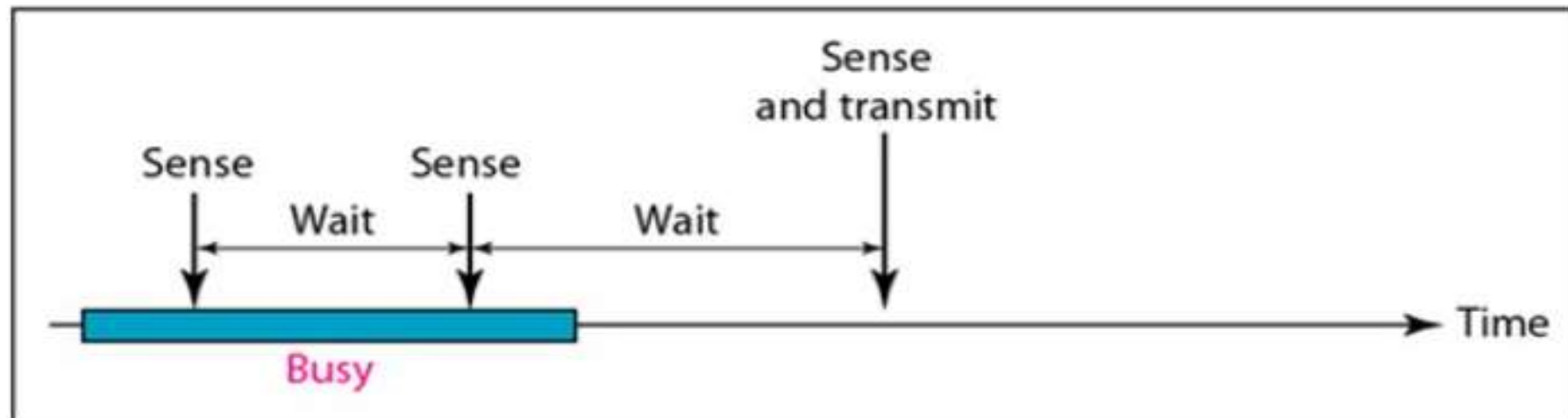
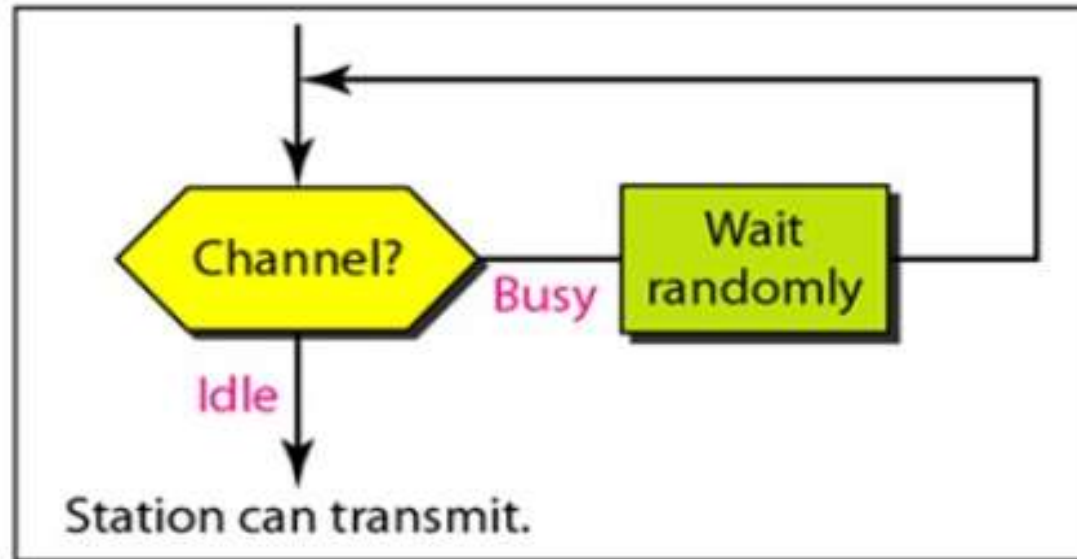
1-persistent CSMA



nonpersistent CSMA

- When a station has a frame to send, it first listens to the channel (idle or busy)
 - If the channel is idle, the station sends its data.
 - if the channel is busy, the station waits a random period of time and then repeats the algorithm
 - this algorithm leads to better channel utilization but longer delays than 1-persistent CSMA.

nonpersistent CSMA



CSMA with Collision Detection (CSMA/CD)

- In this protocol, the station listen to the channel while it is transmitting, and when it detects the collision, it stops transmitting to save time and bandwidth.
- This protocol is the basis of the Ethernet LAN.

4.2.3 Collision-Free Protocols

- there will never be any collisions (0% collision)
- Versions of Collision-Free Protocols
 - A Bit-Map Protocol
 - Token Passing
 - Binary Countdown

A Bit-Map Protocol

- (before any transmission happens) there is contention period consists of number of slots, each slot is assigned to a station.
- Each station (has a frame to be sent) sends a 1 bit during the its slot
- At the end of the contention period the order of the transmissions (that are going to happen) are known

Token Passing protocol

- **Token Passing** is to pass a small message called a token from one station to the next in the same predefined order. (The token represents permission to send).
- If a station has a frame queued for transmission when it receives the token, it can send that frame before it passes the token to the next station.
- If it has no queued frame, it simply passes the token.

Token ring protocol

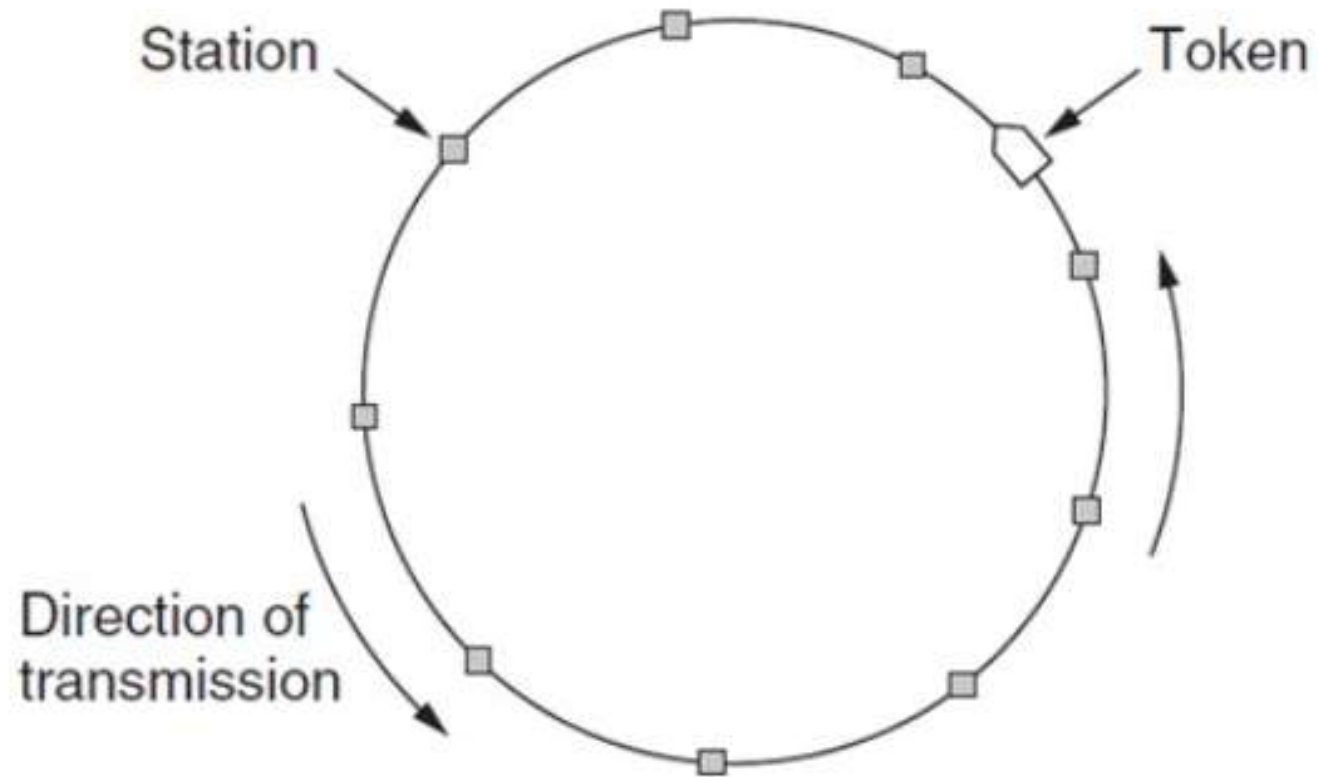


Figure 4-7. Token ring.



Binary Countdown

- A station wanting to use the channel broadcasts its address as a binary bit string, starting with the high order bit.
- The bits in each address position from different stations are BOOLEAN ORed together by the channel when they are sent at the same time.
- The higher-numbered stations have a higher priority than lower-numbered stations.

Binary Countdown

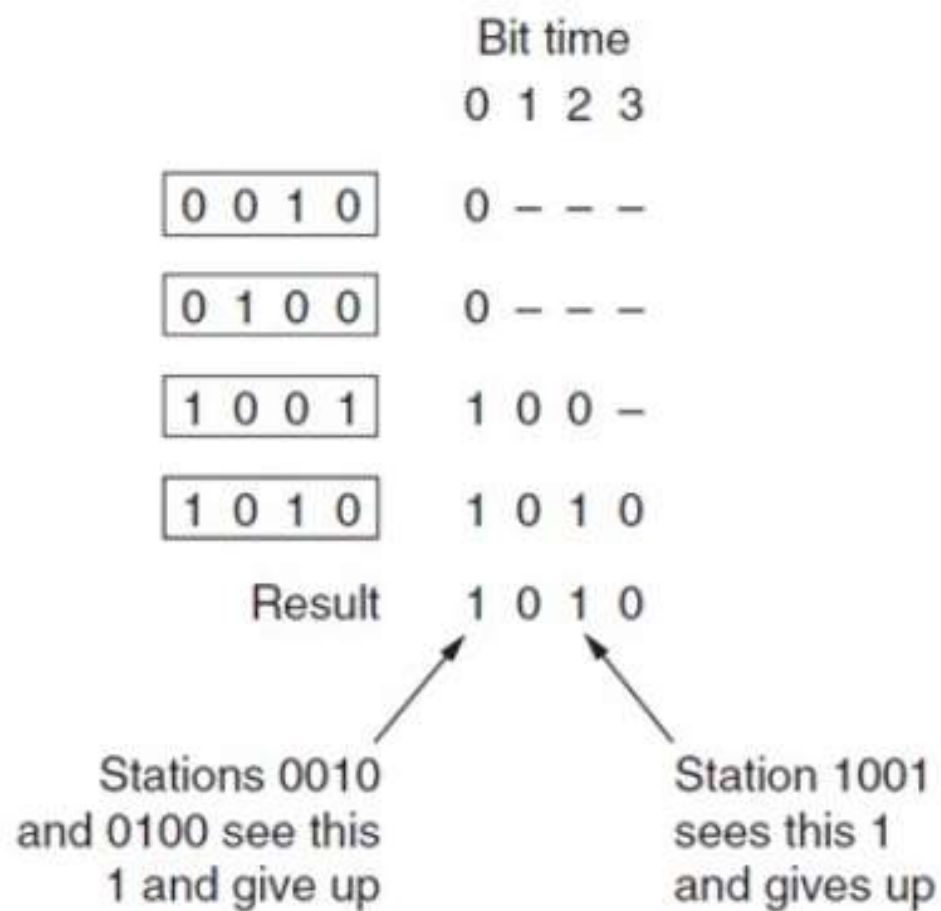
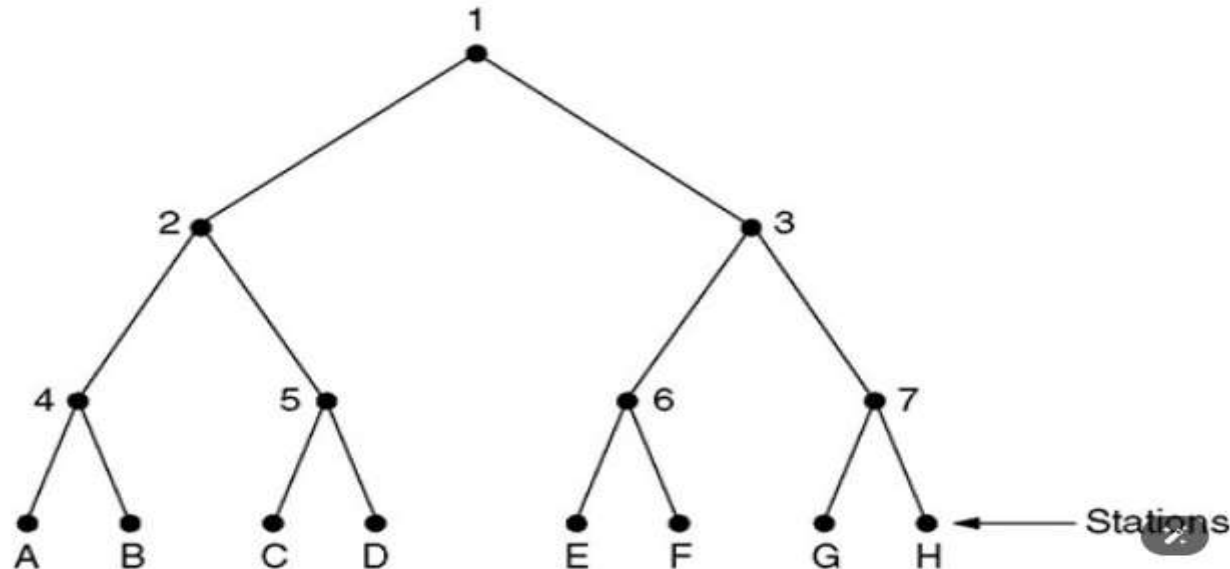


Figure 4-8. The binary countdown protocol. A dash indicates silence.

4.2.4 Limited-Contention Protocols

- **limited-contention protocols** : It combines the best properties of the contention protocols (ALOHA and CSMA) and collision-free protocols
- e.g. The Adaptive Tree Walk Protocol

The Adaptive Tree Walk Protocol



4.2.5 Wireless LAN Protocols

- using CSMA in Wireless LAN leads to 2 problems:
 - The hidden terminal problem: happens when a station not being able to detect a potential competitor for the medium because the competitor is too far away.
 - The exposed terminal problem: happens when a station falsely concludes that it may not start transmission while it is possible.

The hidden terminal problem

A wants to send to B
but cannot hear that
B is busy

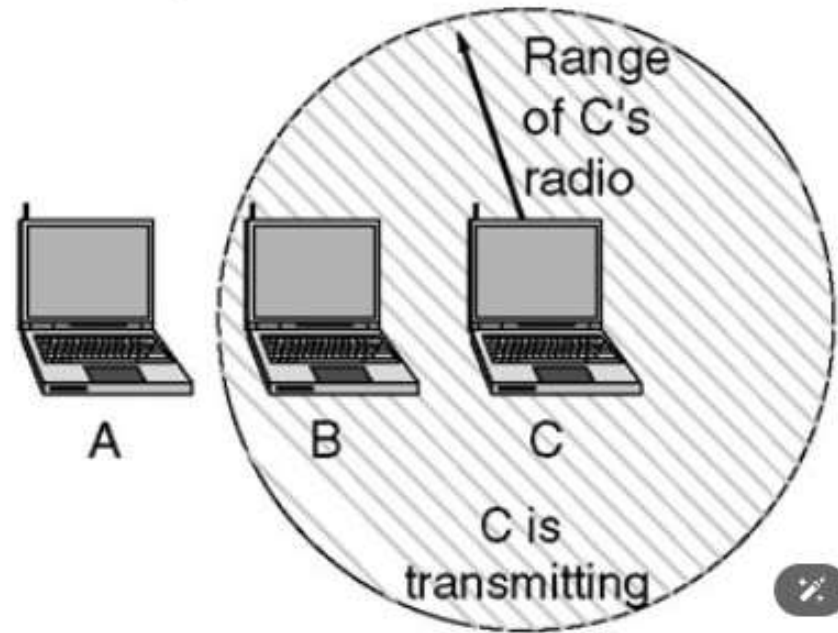


Figure 4-11. A wireless LAN. (a) A and C are hidden terminals when transmitting to B.

The exposed terminal problem

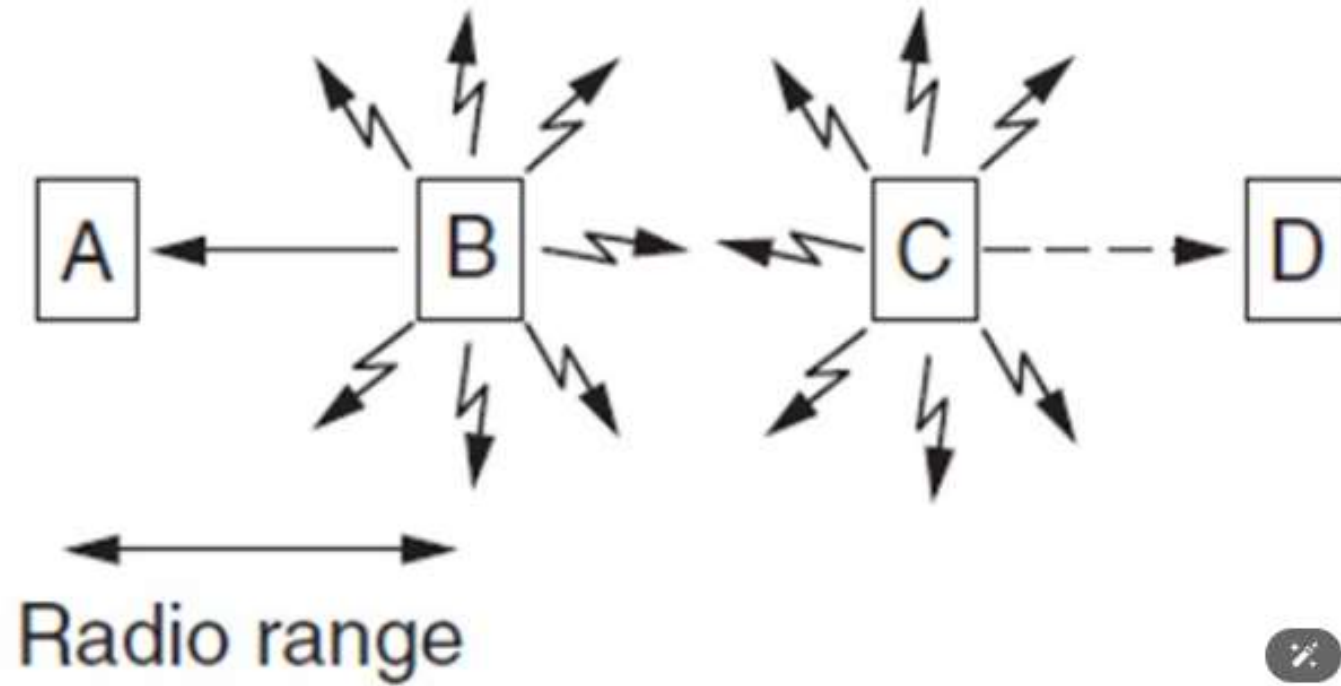
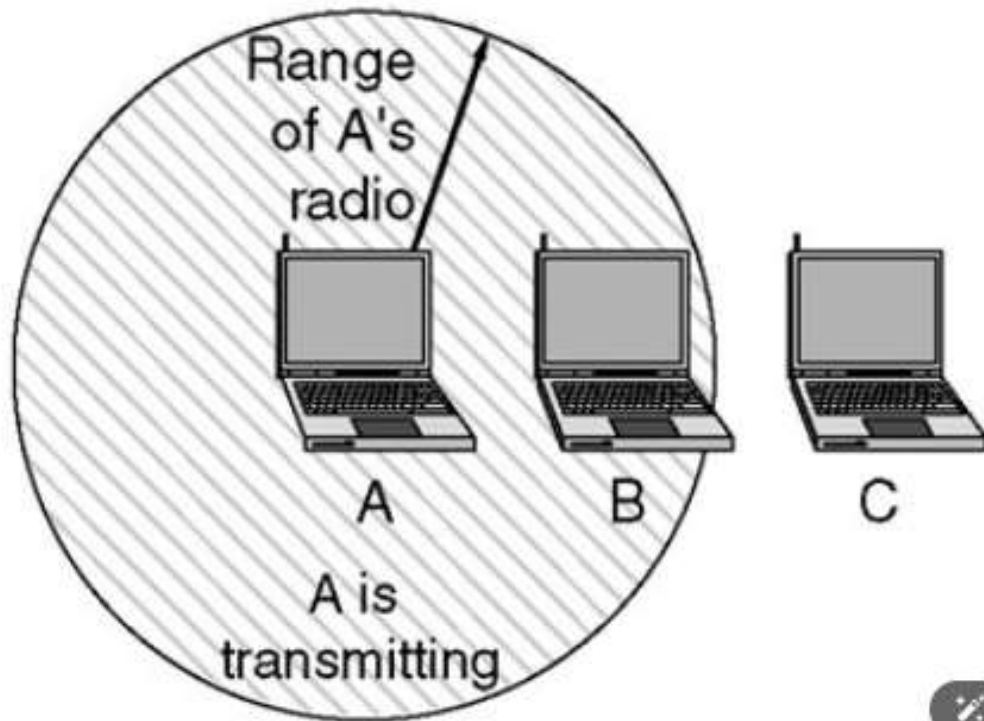


Figure 4-11. A wireless LAN. (b) *B* and *C* are exposed terminals when transmitting to *A* and *D*.

The exposed terminal problem (1)

B wants to send to C
but mistakenly thinks
the transmission will fail



4.2.5 Wireless LAN Protocols : MACA

- **MACA** (Multiple Access with Collision Avoidance) : The basic idea behind it is for the sender to stimulate the receiver into outputting a short frame, so stations nearby can detect this transmission and avoid transmitting for the duration of the upcoming (large) data frame.

Multiple Access with Collision Avoidance (MACA)

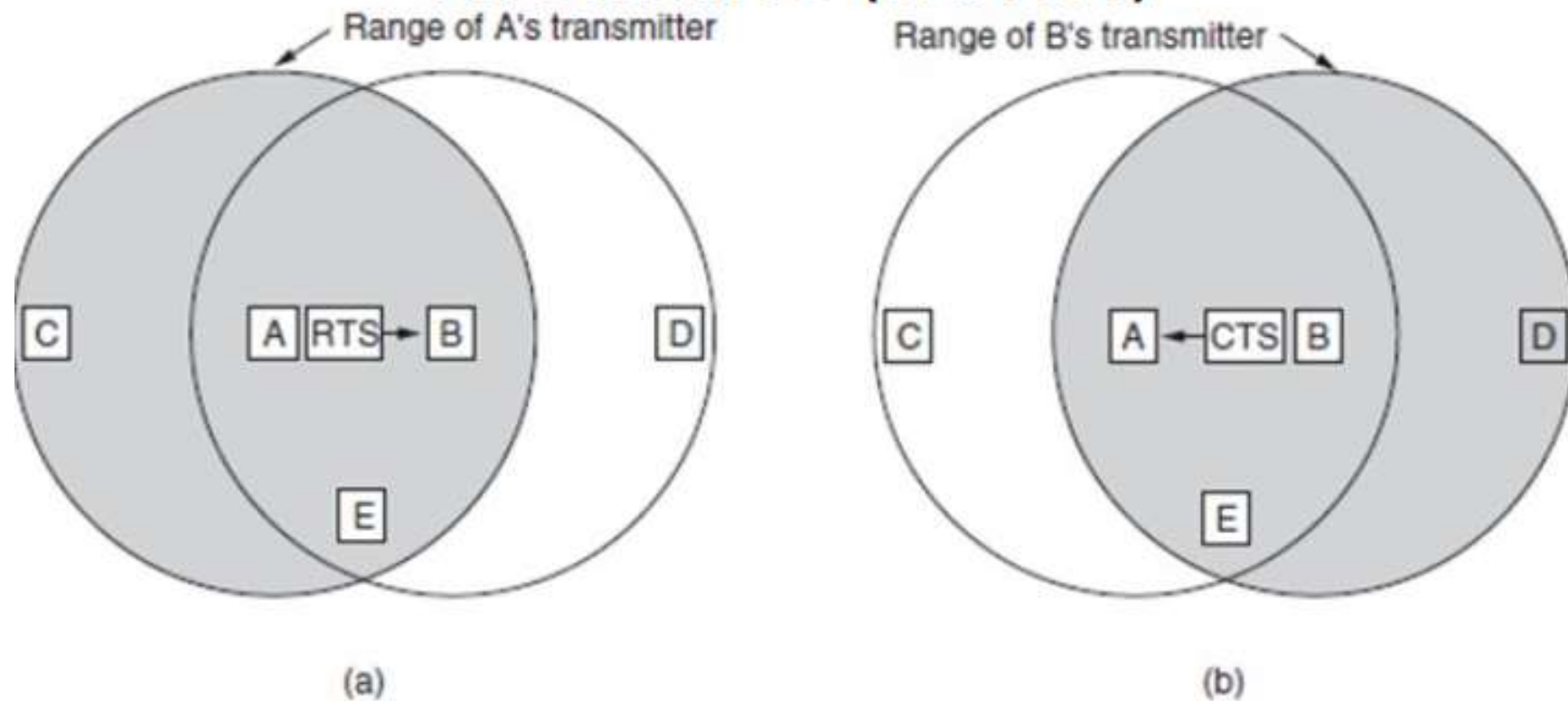


Figure 4-12. The MACA protocol. (a) A sending an RTS to B. (b) B responding with a CTS to A.

